

Lasso Trading

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1 Introduction

The volume of data available worldwide are growing at an unprecedented pace. In 2024 alone, the world generated approximately 402 million terabytes of data per day, up from 43 million just a decade earlier (Wright, 2025). A consequence of this data explosion is that investors today can base their decisions not only on detailed business data but also on a wide array of alternative data sources such as social media activity, satellite images, and web traffic. This data is in high demand with investors. In a survey among 150 asset-managers, 91% indicated that they already use, or plan to implement, machine learning and artificial intelligence in their investment-process (Mercer, 2024). In the standard learning-based asset-pricing model, agents are endowed with a model that relates future returns to a small set of parameters (Timmermann, 1993; Hong, Stein and Yu, 2007). Agents use observed data to estimate these parameters and form return expectations. Evidently, this is a poor representation of how investors form return expectations in a data-rich environment.

We address this gap by studying a learning mechanism that allows agents to search a large number of potential predictors for return predictability and to update the set of predictors they rely on over time. Specifically, we ask: *Can the sparse and unstable nature of return predictability be explained as the outcome of investors actively searching for it in a large number of potential predictors?*

This paper contributes in two ways. First, we bring theory closer to how investors report forming expectations today: by combining many potential predictors using machine learning. Second, our framework is well suited to match a central empirical pattern in return predictability: it is sparse and unstable (Chinco, Clark-Joseph and Ye, 2019; Farmer, Schmidt and Timmermann, 2023). At any point in time, only a small subset of signals forecasts returns, and the elements of this subset change over time.

The central novelty of our model is a learning mechanism based on the least absolute shrinkage and selection operator (LASSO). LASSO is a regularized regression method that selects the most relevant predictors from a large set of candidates by shrinking small coefficients to zero (Tibshirani, 1996). In the model, investors do not possess a structural theory

linking fundamentals to returns but rather belief that returns may be predictable from a large set of candidate predictors. They use a simple linear forecasting rule that they estimate using LASSO. The central mechanism in the model is a feedback loop from learning to prices: the act of searching for return predictability itself generates return predictability. The economy endogenously generates *sparse and short-lived episodes of return predictability*, despite the absence of predictability under rational expectations.

Related Literature

The paper relates to three strands of research.

Learning in asset pricing. A large literature studies bounded rationality by assuming that agents learn unknown parameters of a correctly specified model using least squares methods (Marcet and Sargent, 1989). Classic results show that learning can generate excess volatility and return predictability (Timmermann, 1993, 1996). Subsequent work emphasizes that convergence to rational expectations need not obtain when agents have limited memory or employ misspecified forecasting models (Honkapohja and Mitra, 2003). Related contributions allow agents to switch among multiple forecasting rules or to rely on parsimonious PLMs in multivariate environments (Barberis, Shleifer and Vishny, 1998; Hong, Stein and Yu, 2007; Branch and Evans, 2010).

Internal rationality and belief–price feedback. A central challenge in standard learning models is that agents often know the mapping from fundamentals to prices. Following Adam, Marcet and Nicolini (2016), we adopt an internal rationality perspective in which agents optimize conditional on subjective beliefs about returns and learn from realized prices. This framework generates a feedback loop between beliefs and equilibrium outcomes. We retain the analytical discipline of a Lucas-tree economy (Lucas, 1978), while replacing ad hoc learning rules with a microfounded, empirically prevalent high-dimensional procedure.

Sparse and unstable predictability. Empirical evidence documents that predictive relations are time-varying (Ang and Bekaert, 2007; Goyal and Welch, 2003; Lettau and Ludvigson, 2001) and sparse across a large set of candidate signals (Kozak, Nagel and Santosh, 2020; Freyberger, Neuhierl and Weber, 2020; Feng, Giglio and Xiu, 2020). Machine-learning methods exploit this sparsity and often outperform traditional linear models (Gu, Kelly

and Xiu, 2020). Most closely related, [Chinco, Clark-Joseph and Ye \(2019\)](#) show that LASSO uncovers high-frequency predictability that is unstable and concentrated in a small set of predictors. Our contribution is to endogenize these features as equilibrium outcomes of learning and trading.

Model

Environment. We study a standard [Lucas \(1978\)](#) asset-pricing model in the flavour of [Adam, Marcet and Nicolini \(2016\)](#). A representative risky asset pays dividend D_t and its price P_t satisfies the Euler equation

$$P_t = \delta \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} (P_{t+1} + D_{t+1}) \right], \quad (1)$$

where preferences exhibit constant relative risk aversion with coefficient γ and discount factor δ . Under rational expectations, consumption and dividends follow stationary log-normal processes and returns are conditionally unpredictable.

Perceived law of motion. Following [Adam, Marcet and Nicolini \(2016\)](#) we assume that agents are fully rational with respect to the dividend and consumption processes but do not with respect to returns. Instead, they believe that returns are predictable conditionally on one or multiple signals. We formalize this by endowing agents with a perceived law of motion (PLM) for log returns of the form

$$r_{t+1} = X_t' \beta + u_{t+1}, \quad (2)$$

where X_t is a potentially high-dimensional vector of predictors. Agents try to estimate the unknown coefficient vector β using observed data.

Actual law of motion. The agents' expectations of future returns $X_t' \beta$ affect actual returns through the Euler equation. Substituting the agents' subjective beliefs into [1](#), we obtain the ALM for returns:

$$r_{t+1} = \log(\varepsilon_{t+1}) + \log(1 - \kappa e^{x_t \beta}) - \log(1 - \kappa e^{x_{t+1} \beta}), \quad (3)$$

where ε_{t+1} is an i.i.d. innovation and $\kappa := \delta a^{-\gamma} e^{\sigma_u^2/2}$. Note, that the resulting ALM is nonlinear, implying that the agents' linear PLM is misspecified.

Equilibrium. Under a misspecified PLM, learning does not necessarily converge to the rational expectations equilibrium over time (Honkapohja and Mitra, 2003). Instead, we focus on a notion of equilibrium that requires internal statistical consistency between beliefs and outcomes. We adopt a cross-moment consistent equilibrium (CMCE), in which the joint distribution of returns and predictors induced by the ALM is such that, when agents apply their learning rule (e.g.: OLS or LASSO) to this distribution, they recover the same coefficients they use to form beliefs. In a CMCE, prices are determined optimally given subjective beliefs, beliefs are updated using observed data generated by equilibrium prices, and the perceived forecasting moments coincide with the corresponding moments implied by the actual law of motion.

LASSO Learning. We endow our agents with a LASSO learning rule. In each period, they use LASSO regression on a rolling window of recent data to estimate the parameters of their PLM. Agents update beliefs about return predictability using LASSO on a short, rolling sample of recent data. Formally, agents choose the vector β by solving the penalized least-squares problem

$$\hat{\beta}_t \in \arg \min_{\beta} \left\{ \frac{1}{2} \sum_{i=t-n}^{t-1} (r_{i+1} - X_i' \beta)^2 + \lambda \|\beta\|_1 \right\}, \quad (4)$$

where n is the window length and λ is the ℓ_1 -penalty. The penalization shrinks many coefficients to exactly zero which allows the agent to consider more parameters k than there are observations n available. In the case of orthogonal regressors, we can derive a closed form solution that reduces to soft-thresholding of the least-squares estimate,

$$\hat{\beta}_{\text{LASSO}} = S(\beta_{\text{OLS}}, \lambda) = \text{sign}(\beta_{\text{OLS}}) \cdot \max(0, |\beta_{\text{OLS}}| - \lambda), \quad (5)$$

so a predictor is used only when its recent estimated effect is large enough to survive the penalty.

Main result.

Empirical Strategy

We implement a two-stage empirical design. In the first stage, we estimate rolling LASSO forecasting regressions using a high-dimensional predictor set that includes financial vari-

ables and news-based text topics. In the second stage, we estimate the equilibrium feedback parameter κ from the nonlinear return mapping implied by the model.

The theory yields sharp predictions regarding sparsity, instability, and the conditions under which learning-driven trading has economically meaningful price effects.

Conclusion

This paper provides a tractable equilibrium framework in which high-dimensional, sparsity-inducing learning rules endogenously generate sparse and transient return predictability. The model rationalizes empirical patterns documented in the data and connects modern machine-learning forecasting practices to equilibrium asset pricing.

References

- Adam, Klaus, Albert Marcet, and Juan Pablo Nicolini.** 2016. "Stock Market Volatility and Learning." *Journal of Finance*, 71.
- Ang, Andrew, and Geert Bekaert.** 2007. "Stock Return Predictability: Is It There?" 20(3): 651–707.
- Barberis, Nicholas, Andrei Shleifer, and Robert Vishny.** 1998. "A model of investor sentiment." *Journal of Financial Economics*, 49(3): 307–343.
- Branch, William A., and George W. Evans.** 2010. "Asset return dynamics and learning." *Review of Financial Studies*, 23.
- Chinco, Alex, Adam D. Clark-Joseph, and Mao Ye.** 2019. "Sparse Signals in the Cross-Section of Returns." *Journal of Finance*, 74.
- Farmer, Leland E., Lawrence Schmidt, and Allan Timmermann.** 2023. "Pockets of Predictability." *Journal of Finance*, 78.
- Feng, Guanhao, Stefano Giglio, and Dacheng Xiu.** 2020. "Taming the Factor Zoo: A Test of New Factors." *Journal of Finance*, 75.
- Freyberger, Joachim, Andreas Neuhierl, and Michael Weber.** 2020. "Dissecting characteristics nonparametrically." *The Review of Financial Studies*, 33(5): 2326–2377.
- Goyal, Amit, and Ivo Welch.** 2003. "Predicting the Equity Premium with Dividend Ratios." 49(5): 639–654.
- Gu, Shihao, Bryan Kelly, and Dacheng Xiu.** 2020. "Empirical asset pricing via machine learning." *The Review of Financial Studies*, 33(5): 2223–2273.
- Hong, Harrison, Jeremy C Stein, and Jialin Yu.** 2007. "Simple Forecasts and Paradigm Shifts." *The Journal of Finance*, 62(3): 1207–1242.
- Honkapohja, Seppo, and Kaushik Mitra.** 2003. "Learning with bounded memory in stochastic models." *Journal of Economic Dynamics and Control*, 27.
- Kozak, Serhiy, Stefan Nagel, and Shrihari Santosh.** 2020. "Shrinking the cross-section." *Journal of Financial Economics*, 135(2): 271–292.

- Lettau, Martin, and Sydney Ludvigson.** 2001. "Consumption, Aggregate Wealth, and Expected Stock Returns." 56(3): 815–849.
- Lucas, Robert E.** 1978. "Asset Prices in an Exchange Economy." *Econometrica*, 46.
- Marcet, Albert, and Thomas J. Sargent.** 1989. "Convergence of least squares learning mechanisms in self-referential linear stochastic models." *Journal of Economic Theory*, 48.
- Mercer.** 2024. "Investment Management AI Survey." Mercer. Global survey of asset managers.
- Tibshirani, Robert.** 1996. "Regression Shrinkage and Selection Via the Lasso." *Journal of the Royal Statistical Society. Series B: Methodological*, 58. original lasso paper.
- Timmermann, Allan.** 1996. "Excess Volatility and Predictability of Stock Prices in Autoregressive Dividend Models with Learning." *Review of Economic Studies*, 63.
- Timmermann, Allan G.** 1993. "How learning in financial markets generates excess volatility and predictability in stock prices." *Quarterly Journal of Economics*, 108.
- Relevance:**
- First paper to integrate learning about dividend growth process into asset pricing model
 - uses least squares learning with varying window size
 - estimate nonlinear alm using monte carlo simulation
 - finds that learning makes prices excessively volatile and returns predicable from dividend yield
 - effects become smaller as window size increases.
- Wright, Adam.** 2025. "Worldwide IDC Global DataSphere Forecast, 2025–2029." International Data Corporation (IDC) Market Forecast Report US53363625. Accessed: date you accessed the report.